MoodCapture: Motivation & Problem

Depression Detection Using In-the-Wild Smartphone Images

Traditional Methods

Clinical Interviews & Self-Report Questionnaires

- **High User Burden**
  (e.g., a stopwatch and a person looking)
- **Recall Bias**
  (e.g., a thought contradictory dates or facts)
- **Artificial/Controlled Conditions**
  (e.g., a simple illustration of a clinical observation room or a sterile environment)



MoodCapture Solution

Passive Smartphone Image Capture in Real-World **Environments**

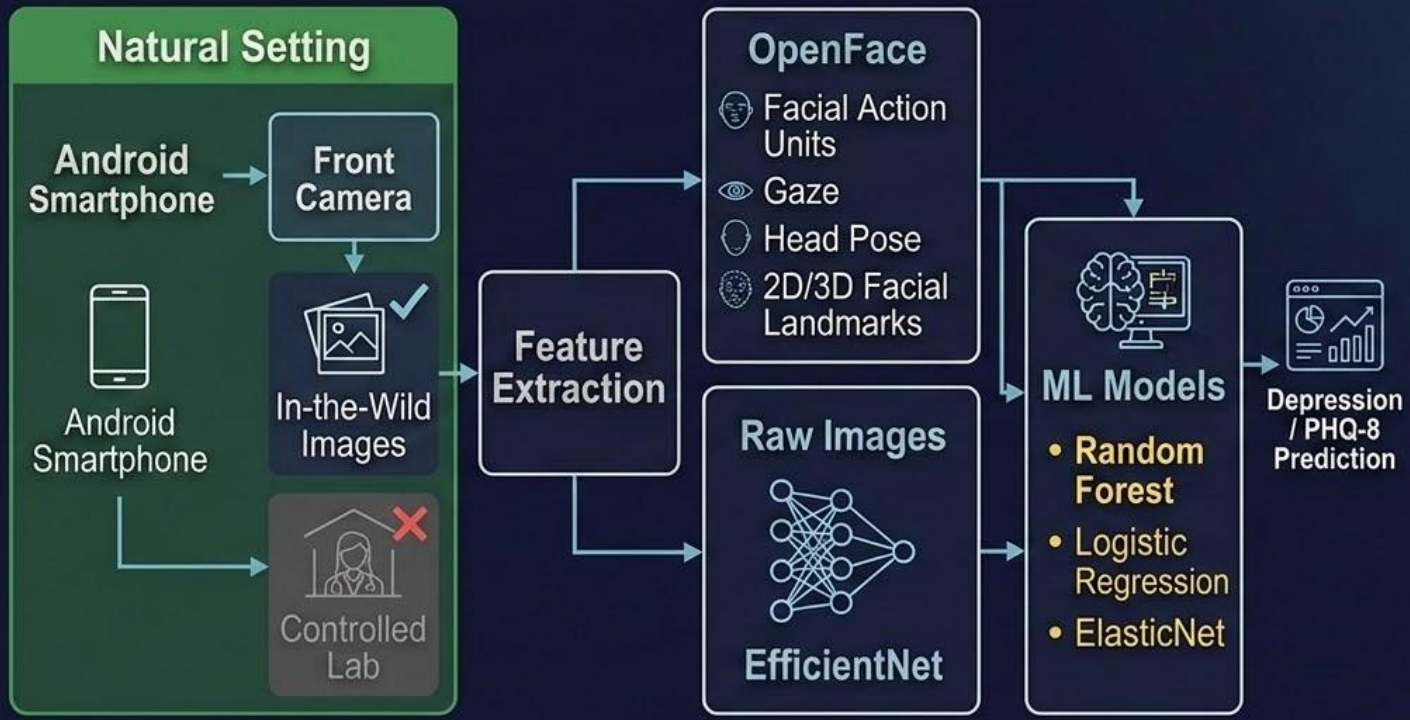
- Passive Capture
- Natural Settings

Traditional (Questionnaire/Clinical Setting) → **MoodCapture** (Natural Smartphone Use)

Paper DOI: 10.32604/cmc.2022.028229 | Computer Modeling in Engineering & Sciences

MoodCapture: System & Methodology

Depression Detection Using In-the-Wild Smartphone Images



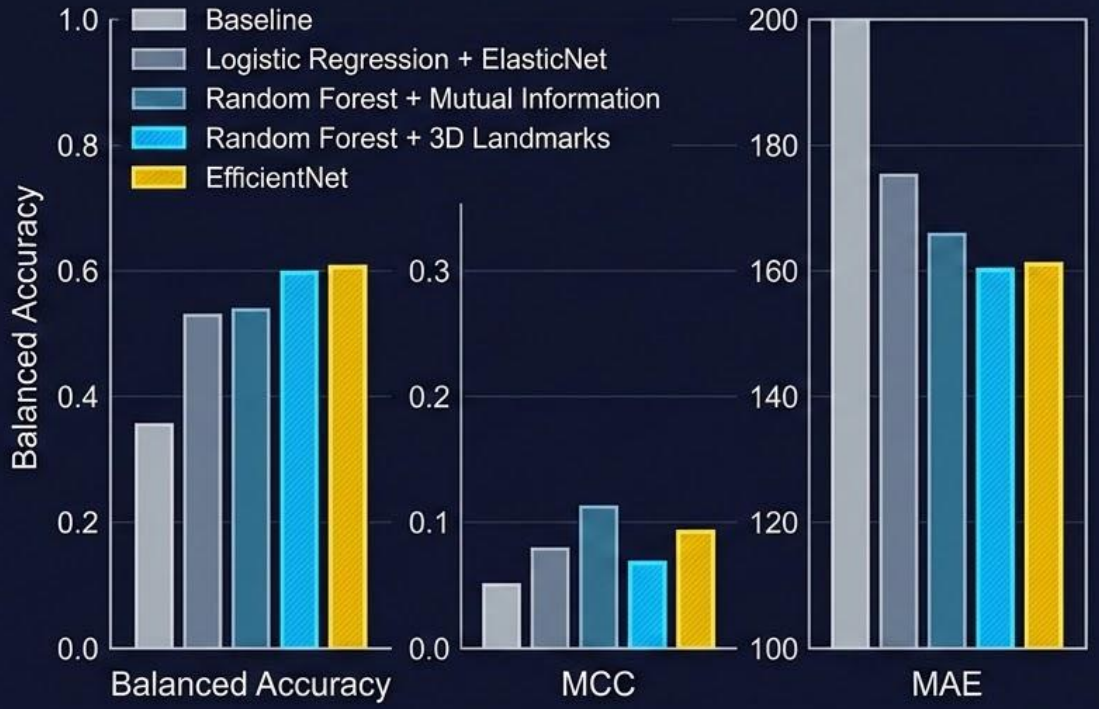
IN-THE-WILD STUDY SETUP

- **90** days
- **177** participants
- **125K+** images
- **87** smartphone models
- PHQ-8 collected **3×** daily

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Results: Can Facial Behavior Predict Depression?

Depression Detection Using In-the-Wild Smartphone Images



Random Forest + 3D facial landmarks:
Balanced Accuracy ≈ 0.60
MCC ≈ 0.14
MAE ≈ 130.31

3D facial landmarks were particularly informative.

EfficientNet:
Balanced Accuracy ≈ 0.61

Balanced Accuracy = classification performance across both classes; MCC = correlation between predictions and ground truth; MAE = PHQ-8 prediction error

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LIMITATIONS, PRIVACY & FUTURE DIRECTIONS

Depression Detection Using In-the-Wild Smartphone Images





1. LIMITATIONS

- Modest Participant Count
- Demographic Imbalance
- Smartphone/Camera Variability
- Relatively Modest Predictive Performance
- Need for Stronger Generalization



2. PRIVACY & ETHICS

- Continuous collection creates serious privacy concerns
- Mental-health information is highly sensitive
- User acceptance is an important HCI challenge



3. FUTURE WORK

- Larger & More Diverse Datasets
- On-device Inference
- Federated Learning
- Privacy-Preserving Processing
- Improved Deep-Learning Models

MoodCapture demonstrates the feasibility of unobtrusive, real-world depression monitoring—but privacy, bias, and clinical reliability remain critical challenges.

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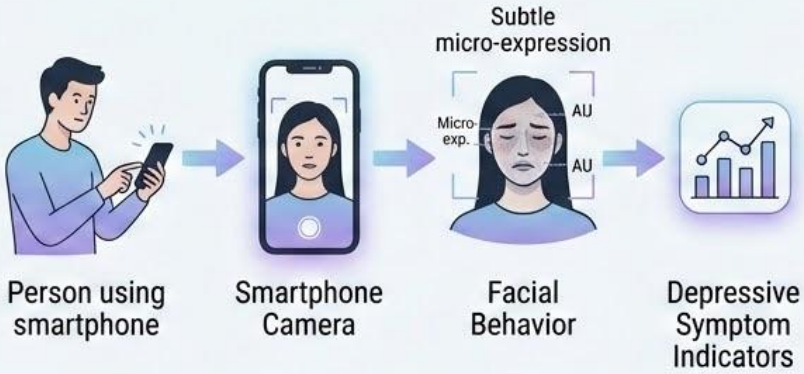
FACEPSY: MOTIVATION & RESEARCH PROBLEM

THE CLINICAL CHALLENGE



- **Traditional Assessments are Subjective & Infrequent**
  - Rely on Patient Self-Reports (e.g., PHQ-9)
  - Conducted in Clinical/ Laboratory Environments
  - Provide only periodic snapshots

THE OBJECTIVE, SENSING-BASED SOLUTION



CONCEPTUAL OVERVIEW

AFFECTIVE COMPUTING: Teaching computers to recognize & process emotion.

MOBILE SENSING: Collecting high-fidelity data via smartphone sensors (e.g., camera).

NATURALISTIC SENSING: Data capture in everyday, uncontrolled environments.

RESEARCH OBJECTIVE: This study investigates whether automatically captured facial behavior can predict depressive symptoms. (N=25 participants)

FacePsy: Smartphone-Based Facial Behavior Sensing

1. Smartphone Camera (Passive Sensing)



Smartphone Camera (Passive Sensing)

2. Face Detection & Tracking (MTCNN)



Real-time face location and tracking in diverse orientations.

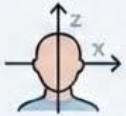
3. Facial Behavior Extraction



Facial Landmarks



Eye-Open State



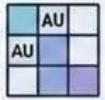
Head Pose



Head Gestures

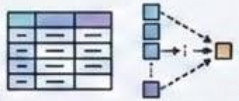


Facial Expressions



Facial Action Units (AUs)

4. Behavioral Features

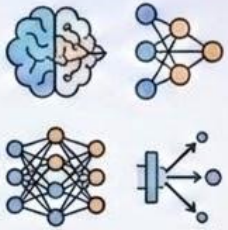


Feature Engineering



Time-Series Fusion

5. Machine Learning (SVM/DNN)



Classification

Supporting Mobile-Sensing Components



Screen/App Triggers
Trigger data collection during natural usage



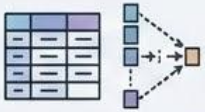
EMA (Ecological Momentary Assessment)
Short, context-aware user surveys



SYSTEM DESCRIPTION: A comprehensive, real-time facial behavior tracking system designed for naturalistic smartphone usage. This architecture passively extracts high-fidelity biomarkers while ensuring low computational overhead for mobile devices.

Prediction Pipeline & Experimental Evaluation

CLASSIFICATION BRANCH
(Depressive Episode Detection)



Facial Behavioral Features (e.g., AUs, Head Pose, Landmarks)



Classification

Prediction:
Depressive Episode Categories (Depressed / Not Depressed)

EVALUATION: AUROC (Area Under ROC Curve)

PHQ-9 (Patient Health Questionnaire-9): A standardized 9-item clinical questionnaire for screening, diagnosing, and **monsification:** Predicts discrete categorical outcomes (e.g., presence or absence of a depressive episode).
Regression: Predicts a continuous numerical value (e.g., exact PHQ-9 score) based on input data.

REGRESSION BRANCH
(PHQ-9 Score Prediction)



Facial Behavioral Features (e.g., AUs, Head Pose, Landmarks)

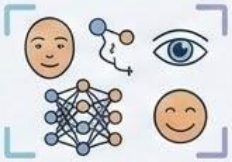


Regression

Prediction: Continuous Numerical PHQ-9 Score

EVALUATION: MAE (Mean Absolute Error)

Conceptual ML Overview





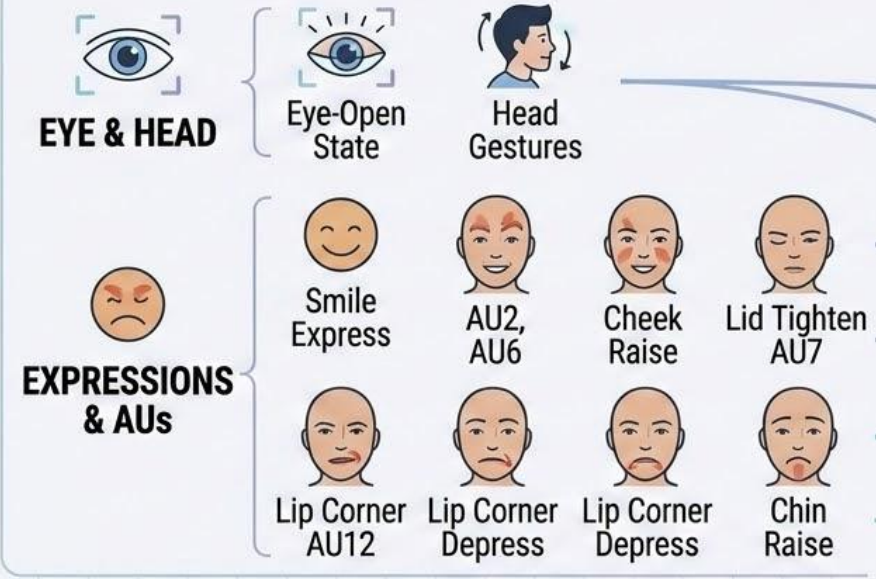
Machine Learning model



Based on methodology referenced in Computer Modeling in Engineering & Sciences (DOI: 10.32604/cmc.2022.028229)

Results & Key Findings

Feature Impact on Classification & Regression



AUROC ≈ 0.81

≈ 0.81

ROC

Depressive Episode Classification

LIMITATION: Small Sample Size (25 participants) → Further Validation Required

MAE = 3.08

= 3.08

PHQ-9 Symptom Score Regression

SMARTPHONE FACIAL SENSING can capture meaningful behavioral signals.

MULTIPLE FINE-GRAINED FACIAL BEHAVIORS are associated with depressive symptoms.

FACEPSY DEMONSTRATES THE POTENTIAL of naturalistic mobile sensing for mental-health research.